THE MASTERMIND COACHING CLASSES

Revision Notes

Class 6 · Science

NCERT Curiosity — Chapters 1 to 6 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
1	The Wonderful World of Science	Curiosity, scientific method, science everywhere
2	Diversity in the Living World	Biodiversity, grouping, adaptation, habitat
3	Mindful Eating: A Path to a Healthy Body	Nutrients, balanced diet, food tests, millets, food miles
4	Exploring Magnets	Magnetic materials, poles, compass, attraction and repulsion
5	Measurement of Length and Motion	Units, SI system, correct measurement, reference point, types of motion
6	Materials Around Us	Properties of materials, classification, matter, mass and volume

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CHAPTER 01

The Wonderful World of Science

This opening chapter welcomes you to Science as a way of thinking, observing and doing that helps us understand the world and uncover the secrets of the universe. It shows that science grows out of curiosity, that everyone uses the scientific method in daily life, and previews the questions the book will explore.

Big Questions

- What is Science and why is curiosity its starting point?
- What steps make up the scientific method?
- Where do we meet science in everyday life?

1.1 What Is Science?

- **Science** is a way of *thinking, observing and doing things* to understand the world we live in and to uncover the secrets of the universe.
- It is like a big adventure: we **ask questions**, explore the world and try to understand how things work.
- The most important quality for a learner of science is **curiosity** — this is why the book is titled *Curiosity*.
- Science studies everything: tiny grains of sand and massive mountains, a leaf of grass and a vast forest, the stars in the sky and the flower that blooms.
- Science is **everywhere** — in mountainous regions, deserts, coasts, the depths of the ocean, galaxies in outer space, the kitchen and the playground. Some of the most important discoveries have come from unexpected places.

REMEMBER

Science is **not** just memorising facts or doing experiments. It is a step-by-step process that helps us find answers to our questions.

1.2 Science as a Jigsaw Puzzle

The textbook compares science to a **giant, unending jigsaw puzzle**.

- Every new discovery adds another piece to the puzzle.
- There is no limit to what we can discover, because each new piece of knowledge raises more questions.
- Sometimes a piece has been placed wrongly and must be moved — **new discoveries can change our understanding** of the world.
- Ideas in science are all **connected** with one another; the more we learn, the more we notice these links.

1.3 What Will We Explore in This Book?

The book moves outward from our own home to the wider universe, raising questions at every step.

Theme	Questions the book raises	Chapter area
Planet Earth and living things	Earth is the only known planet that supports life; how do the plants and animals of different regions survive and grow? How does a seed become a plant or a caterpillar a butterfly?	Diversity in the living world

Theme	Questions the book raises	Chapter area
Food	India has many cuisines; what are dishes made of and how do we find out?	Mindful eating
Water	Why does it rain? Why does water freeze to ice and boil to steam?	States of water
Hot and cold	How do we understand hot and cold, e.g. in a fever or a drink?	Temperature
Materials	Paper, metal key, plastic ruler, rubber eraser, magnet, clothes: what are they made of and how are mixed materials separated?	Materials and separation
Things beyond Earth	What are the Sun, Moon and millions of stars?	Beyond Earth

Other chapters look at the structure of a leaf, how things move, and even separating the skin of a peanut seed.

1.4 The Scientific Method

The textbook introduces the method with a familiar problem — **your pen stops writing**.

Step 1 — Observe and wonder
You notice something interesting or puzzling (the pen does not write) and ask a question: *Why did my pen stop writing?*

Step 2 — Guess a possible answer
You guess that the ink has finished (a guess that can be tested is called a hypothesis).

Step 3 — Test the guess
You open the pen and check the refill. If it is empty, the guess was correct.

Step 4 — Analyse the results
If the refill still has ink, the guess was wrong, so you make another guess (perhaps the ink has dried up) and test that as well.

IMPORTANT

The scientific method: **observe → ask a question → guess an answer → test the guess (experiments / more observations) → analyse the results**. Scientists follow this method, but *anyone who follows it is working like a scientist*.

1.5 Everyone Is a Scientist at Times

Person	Question they ask	What they are doing
A cook	Why has the dal spilled out of the cooker — was there too much water?	Guessing a cause and testing it next time
A bicycle repair person	Where has the air leaked from the flat tyre?	Observing and testing to locate the fault
An electrician	Is the problem in the bulb or in the switch?	Testing one guess after another

So in daily life we all use the scientific method, knowingly or unknowingly. Studying science makes us better at solving **bigger problems** and solving more mysteries of the universe.

1.6 Habits of a Good Learner of Science

- Be **curious** and **observe** your surroundings keenly.
- Keep asking *how* and *why*; the world is full of things waiting to be explored.
- Science is rarely done alone — scientists across the world **work together** in large teams, so ask friends for help and discover things together.
- You will not find answers to all questions in Grade 6; it is a journey of the next five years and beyond.
- Enjoy the scientific journey — like children enjoying the rain, science is *joyful exploration*.

1.7 Activities in This Chapter

Activity	What you are asked to do
1.1	Write about a similar problem you solved and the steps you took
1.2	Describe a daily-life situation in which someone followed a scientific method
1.3	Choose something you would ask “Why?” about and write how you would try to find the answer

1.8 Quick Revision Questions

Likely question	Model answer
What is science?	A way of thinking, observing and doing things to understand the world and uncover the secrets of the universe.
Why is the book named *Curiosity*?	Curiosity is the most important quality for learning science; it makes us ask questions.
Why is science compared to a jigsaw puzzle?	Every discovery adds a piece, new questions never end, and a wrongly placed piece may have to be moved when understanding changes.
List the steps of the scientific method.	Observe and wonder, ask a question, guess an answer, test the guess, analyse the results.
Give an example of the scientific method from daily life.	Finding why a pen stopped writing: guess empty ink, check the refill, then try another guess if ink is present.
Who is a scientist?	Anyone who follows the scientific method to solve problems or discover new things.
Why should scientists work together?	Science is rarely done alone; large teams across the world share ideas and solve harder problems.
Is science only memorising facts?	No. It is a step-by-step process of finding answers to questions.
Which planet is the only one known to support life?	Earth.
How can discoveries change science?	New discoveries may show that an earlier idea was wrong and change our understanding of the world.

★ KEY TAKEAWAYS

- **Science** is a way of thinking, observing and doing; its starting point is **curiosity**.
- Science is everywhere — from sand grains to galaxies, kitchen to playground.
- Scientific method: **observe, question, guess, test, analyse**.
- Everyone who follows the method works like a scientist (cook, mechanic, electrician).
- Knowledge grows like a **jigsaw puzzle**; new discoveries can change earlier ideas.
- Scientists work **together**; keen observation and asking *why* are habits to build.

CHAPTER 02

Diversity in the Living World

*A nature walk with a scientist (Dr Raghu) and bird-call expert Maniram chacha opens the chapter and shows the huge **variety of plants and animals** around us. We learn to observe them, group them by common features, understand why different regions have different life (adaptation, habitat) and why protecting biodiversity matters.*

Big Questions

- How do plants and animals around us differ from one another?
- On what basis can we group them?
- Why do plants and animals of different regions look different?
- Why must we protect biodiversity?

2.1 Diversity in Plants and Animals Around Us

On the nature walk the students are told to carry a notebook, pen and water bottle, to notice smells and sounds, to **respect all living creatures and observe without disturbing** them (do not pluck leaves or flowers). They see grasses, bushes, large trees, birds, butterflies and monkeys. Each bird has its own chirp — an example of **diversity** in nature.

▸ Observing plants (Activity 2.1, Table 2.1)

Plants differ in stem, leaves and flowers. Features to record: height (tall/short), stem (hard/soft, thin/thick), leaf shape and arrangement, flower colour, shape and scent.

Plant	Stem	Leaves	Flowers / notes
Common grass	Soft and thin	Green; a single leaf grows alternately from different points on the stem	—
Tulsi (holy basil)	Hard and thin	Pairs of leaves arranged in opposite directions	Pinkish purple
Hibiscus (gudhal)	Hard	Leaves with a smooth surface	Showy flowers
Neem	Hard, thick	—	—

▸ Observing animals (Table 2.2)

Animals differ in where they live, what they eat and how they move. Examples: **crow** (lives on trees, eats insects, flies and walks), **ant** (nests in soil, six legs, eats leaves, seeds, insects), **cow** (eats grasses and leaves). Some animals live on land, some on trees (birds), fish live in water, and frogs live on land as well as in water. Animals eat very different foods and move in different ways.

▸ Biodiversity

> **DEFINITION**
> The **variety of plants and animals found in a particular region** contributes to the **biodiversity** of that region. Each member has a role: trees give food and shelter to birds and other animals; animals spread seeds after eating fruits. Hence plants and animals are **dependent on each other**.

Activity 2.2 (Let us appreciate): each student draws a plant and an animal on the board; the many different drawings show that there are far more varieties than those drawn.

2.2 How to Group Plants and Animals?

Just as grouping books helps organise a school bag, **grouping** (arranging things into groups based on common features) makes it **easier to understand and study** plants and animals by their similarities and differences. Different groups of students may use different bases (height of plants, presence or absence of flowers, hard/soft stem; for animals — food, habitat, colour, movement) — all are valid; the basis chosen simply differs.

▸ 2.2.1 Grouping plants

▸ (a) Herbs, shrubs and trees — on height, stem and branching (Table 2.3)

Group	Height	Stem	Branching	Example
Tree	Very tall	Hard, thick, brown, woody	Branches start higher up on the stem, away from the ground	Mango
Shrub	Not as tall as trees	Many brown woody stems, hard but not as thick	Branches start close to the ground	Rose
Herb	Small / short	Soft, green, tender stem	Branch near the ground	Tomato

- **Climbers** — weak stems that need support to climb.
- **Creepers** — stems that creep along the ground.

▸ (b) Leaf venation

The thin lines on a leaf are **veins**; the pattern of veins is **venation**.

Type	Pattern	Examples
Reticulate venation	Net-like pattern of veins on both sides of a thick middle vein	Hibiscus, sadabahar, chickpea
Parallel venation	Veins run parallel to each other	Banana, grasses, lemongrass, wheat

▸ (c) Roots (Activity 2.6)

Dig small herbs gently with a **khurpi** after wetting the soil, wash the roots, observe and then **replant** them.

Type	Description	Examples
Taproot	One main root with small side roots arising from it	Mustard, hibiscus, sadabahar, chickpea

Type	Description	Examples
Fibrous roots	A bunch of similar-sized thin roots arising from the base of the stem	Grass, lemongrass, wheat

▸ (d) Seeds — monocots and dicots (Activity 2.8)

Soak chickpea and maize seeds for 2–3 days, remove the seed coat of chickpea and compare. A chickpea seed splits into **two parts, each called a cotyledon**; maize has a **single** thin cotyledon.

Feature	Dicotyledons (dicots)	Monocotyledons (monocots)
Cotyledons in seed	Two	One
Leaf venation	Reticulate	Parallel
Root system	Taproot	Fibrous
Examples	Chickpea (chana), kidney beans, sadabahar, hibiscus	Maize, wheat, grass, lemongrass

> **IMPORTANT**
> Generally, plants with **reticulate** venation have **taproots** and are **dicots**; plants with **parallel** venation have **fibrous roots** and are **monocots** (Table 2.4 / Activity 2.7).

▸ 2.2.2 Grouping animals

Animals can be grouped by what they eat, where they live, their colour, shape or size and — as in Activity 2.9 — by **type of movement and the body part used**.

Animal	Type of movement	Body parts used
Ant	Walks, crawls	Legs
Goat	Walks and jumps	Legs
Pigeon	Walks and flies	Legs and wings
Housefly	Walks and flies	Legs and wings
Fish	Swims	Fins

Animals can fly, run, crawl, walk, hop, jump or swim, using wings, legs, fins and other parts.

▸ Know a scientist — Janaki Ammal (1897–1984)

An Indian **botanist** who worked to document and preserve India's plant biodiversity; she played a key role in the **Save Silent Valley** movement and, as head of the **Botanical Survey of India**, started programmes to document India's plant diversity.

▸ Success story — Save Silent Valley movement

- Silent Valley is a **moist evergreen forest in Palakkad district, Kerala**.
- A proposal for a **hydroelectric dam across the Kunthipuzha river** threatened it.
- Common people (not living near the forest) fought the proposal for **10 years** through awareness programmes, letters to editors, newspaper articles, seminars, petitions and court appeals.
- The movement **succeeded** in saving the forest and its biodiversity.

2.3 Plants and Animals in Different Surroundings

Plants and animals found in a desert, mountains, ocean or forest are different (Table 2.6: camel in the desert, deodar on mountains, fish in the ocean, lion in the forest). The biodiversity of a region varies because **conditions differ**.

Region	Conditions	Example and its feature
Hot desert (Rajasthan)	Very little water; very hot by day, cold at night	**Cactus** — thick, fleshy stem stores water
Cold mountains (Himalayas)	Frequent snowfall	**Deodar** — conical shape and flexible, sloping branches let snow slide off
Ocean (Andaman & Nicobar)	Water all around	**Fish** — streamlined body, fins to swim; whales are huge
Shola forests (Nilgiris) / Sikkim	Heavy winds on mountain tops	**Rhododendron** — shorter with smaller leaves in the Nilgiris, taller in Sikkim: even one plant varies by region

▸ Camels of the hot and the cold desert

Feature	Hot desert camel (Rajasthan)	Cold desert camel (Ladakh)
Legs	Long legs with wide hooves — walk on sand without sinking	Comparatively shorter legs — walk easily in mountains
Humps	One hump (stores food for scarcity)	Two humps that shrink in late winter as stored food is used
Other	Excrete little urine, dry dung, do not sweat — survive many days without water	Long hair from head to neck to survive the cold winters

▸ Adaptation and habitat

> DEFINITION
>
> **Adaptations** are the special features that enable plants and animals to survive in a particular region (e.g. shape of deodar, height of rhododendron, streamlined fish body). The **habitat** is the place where a plant or animal lives; it provides **food, water, air, shelter** and other needs. Many types of living things can share a habitat, and habitat shapes biodiversity (sea turtle — ocean; camel — desert; rhododendron — mountains).

▸ Terrestrial and aquatic habitats

Habitat	Where	Examples of habitats	Examples of organisms
Terrestrial (on land)	Land	Forests, deserts, grasslands, mountains	Horse, sheep, squirrel, pigeon, earthworm
Aquatic (in water)	Water	Ponds, lakes, rivers, oceans	Fish, dolphin, whale
Both land and water	—	—	**Amphibians** such as frogs (also crocodile, tortoise in the chapter exercise)

▸ **Know a scientist — Salim Ali (1896–1987)**

The **'Birdman of India'**; travelled across India observing birds, listed them, documented their travel routes and habitats and took steps to conserve regions of high bird diversity — **Keoladeo National Park (Bharatpur, Rajasthan)** and **Ranganathittu Bird Sanctuary (Mandya, Karnataka)**. He wrote a landmark **10-book series** on birds of the Indian Subcontinent and received the **Padma Vibhushan in 1976**.

2.4 Loss of Habitat and Conservation

- Damage to habitats means plants and animals **lose homes, food and other resources**, so **biodiversity is lost** (e.g. a goat without grass, a fish without water).
- Elders may remember birds, insects and animals they saw often in childhood but see rarely now — a sign of damaged habitats.
- Human activities have reduced the **Bengal Tiger, Cheetah and Great Indian Bustard**.

Conservation effort	Year / place	Purpose
Project Tiger	1973	Protect the declining Bengal Tiger
Cheetah Reintroduction Project	2022	Restore the cheetah population
Protected habitats of the Great Indian Bustard	Gujarat, Rajasthan, Maharashtra	Safeguard the bird's habitat

REMEMBER

Sacred groves are undisturbed patches of forest (small to very large) found all over India, protected by the **local community**; no one may harm animals, cut trees or disturb the area. They shelter many plants (including medicinal plants) and animals — a community-protected treasure of biodiversity.

2.5 Keywords

Adaptation, amphibians, aquatic, biodiversity, cotyledon, dicot, monocot, fibrous root, taproot, habitat, herbs, shrubs, tree, parallel venation, reticulate venation, sacred groves, terrestrial, venation.

2.6 Likely Exam Questions

Likely question	Model answer
What is biodiversity?	The variety of plants and animals found in a region.
Why do we group plants and animals?	Grouping on common features makes it easier to understand and study their similarities and differences.
Differentiate between a tree, a shrub and a herb.	Tree: tall, hard thick woody stem, branches high; shrub: shorter, many hard stems branching near the ground; herb: small, soft green stem.
What is venation? Name its two types.	The pattern of veins in a leaf; reticulate (net-like) and parallel.
Compare taproot and fibrous roots with one example each.	Taproot: one main root with side roots (mustard); fibrous: bunch of similar thin roots (grass).
What is the relation between seed type, venation and root?	Dicots: two cotyledons, reticulate venation, taproot; monocots: one cotyledon, parallel venation, fibrous roots.

Likely question	Model answer
What are adaptations? Give two examples.	Features that help survival in a region: fleshy stem of cactus, conical shape of deodar.
What is a habitat? Name two kinds.	The place where an organism lives and gets its needs; terrestrial and aquatic.
How does a camel suit the hot desert?	Long legs with wide hooves, one hump to store food, little urine, dry dung and no sweating.
What are amphibians?	Animals that can live in water as well as on land, e.g. frogs.
How did the Save Silent Valley movement succeed?	People campaigned for 10 years using awareness, letters, articles, seminars, petitions and court appeals against a dam.
Why were sacred groves protected?	Local communities protect these patches, keeping their plants and animals safe.

★ KEY TAKEAWAYS

- **Biodiversity** is the variety of life in a region; organisms depend on one another.
- Plants are grouped as **herbs, shrubs, trees** (also climbers, creepers) and as **monocots / dicots**.
- Dicots: reticulate venation + taproot; monocots: parallel venation + fibrous roots.
- Animals can be grouped by food, habitat, colour or **movement** and body parts used.
- **Adaptations** fit organisms to their **habitat** — terrestrial or aquatic (amphibians use both).
- Habitat damage causes biodiversity loss; Project Tiger (1973), Cheetah project (2022) and sacred groves help conservation.
- Janaki Ammal and Salim Ali worked for plant and bird conservation.

CHAPTER 03

Mindful Eating: A Path to a Healthy Body

*The Sanskrit saying annena jātāni jīvanti means 'food gives life to living beings'. Medu and Mishti learn from nutrition expert Dr Poshita that **'Health is the Ultimate Wealth'**. The chapter studies what we eat, why food habits differ, what nutrients food contains, how to test for them, what a balanced diet is, why millets and local food are good, and why food should not be wasted.*

Big Questions

- Why does food differ from region to region?
- What are the components of food and what do they do?
- How can we test food for starch, fat and protein?
- What is a balanced diet?

3.1 What Do We Eat?

Food is an essential part of daily life. Activity 3.1 asks you to record a week's meals (Table 3.1) and compare with friends: you will find **variety** in what people eat.

▸ 3.1.1 Food in different regions

State	Locally grown crops	Traditional food	Beverages
Punjab	Maize, wheat, chickpea, pulses	Makki di roti, sarson da saag, chhole bhature, parantha, halwa, kheer	Lassi, chhach (buttermilk), milk, tea
Karnataka	Rice, ragi, urad, coconut	Idli, dosa, sambhar, coconut chutney, ragi mudde, palya, rasam	Buttermilk, coffee, tea
Manipur	Rice, bamboo, soya bean	Eromba (chutney), utti (yellow peas curry), singju, kangsoi	Black tea

- Traditional food of a state is usually based on the **crops grown there**.
- India is agricultural with diverse **soil and climate**, so crops differ by region.
- Choice of food also depends on **taste preferences, culture and traditions**.
- Some foods are common to many states; some are eaten only in one.

▸ 3.1.2 How have cooking practices changed?

Cooking practices (**culinary practices**) have changed over time. Earlier cooking used a **chulha** (traditional stove) and grinding was done manually in a **sil-batta** (stone grinder). Today we use a **gas stove** and an **electrical grinder**. Reasons: **technological development, improved transportation and better communication**. Activity 3.3: interview elderly people about food habits, what is new and what caused the changes.

3.2 What Are the Components of Food?

Missing a meal makes us tired and less energetic, so food gives energy. A marathon runner drinks **glucose water** because glucose gives instant energy; glucose is a carbohydrate.

Component	Role	Sources	Group name
Carbohydrates	Primary source of energy (sugar is also a carbohydrate)	Wheat, rice, maize, bajra; potato, sweet potato; banana, mango, pineapple; sugarcane	Energy-giving
Fats	Stored energy; keep us warm (hence winter laddoos with ghee, nuts)	Ghee, butter, oils, groundnut, walnut, almond, coconut, sunflower and pumpkin seeds, soya bean, sesame	Energy-giving
Proteins	Growth and repair; building muscles (sportspersons need more)	Plant: pulses (moong, tuar, gram), kidney beans, peas, soya bean, nuts, edible mushrooms. Animal: milk, paneer, egg, fish, meat	Body-building
Vitamins and minerals	Protect from disease, keep body healthy; needed in small amounts	Fruits, vegetables, milk, whole grains (see table below)	Protective nutrients

More to know: polar bears store fat under the skin to survive months of **hibernation** without eating.

▸ Two cases: deficiency diseases

- **Case 1 — Scurvy:** in 1746 Scottish physician **James Lind** saw that sailors with bleeding, swollen gums recovered after eating lemons and oranges. Scurvy is due to **lack of Vitamin C**.
- **Case 2 — Goitre:** in the 1960s swelling at the front of the neck was common in the Himalayan region and northern plains because the **soil, food and water lacked iodine**. Common salt was supplemented with iodine to make **iodised salt**, which reduced the symptoms.

Agariyas, a tribal community, practise **salt farming** in the Little Rann of Kutch, Gujarat, living about eight months in desert heat.

▸ Vitamins and minerals chart (Fig. 3.5)

Nutrient	Function	Sources	Deficiency disease	Symptoms
Vitamin A	Healthy eyes and skin	Papaya, carrot, mango, milk	Loss of vision	Poor vision, night blindness, sometimes total loss of vision
Vitamin B1	Healthy heart; supports body functions	Legumes, nuts, whole grains, seeds, milk products	Beriberi	Swelling, tingling or burning in hands and feet, trouble breathing
Vitamin C	Helps fight diseases	Amla, guava, green chilli, orange, lemon	Scurvy	Bleeding gums, slow healing of wounds
Vitamin D	Helps absorb calcium for bone and teeth health	Sunlight, milk, butter, fish, eggs	Rickets	Soft and bent bones
Calcium	Keeps bones and teeth healthy	Milk/soya milk, curd, cheese, paneer	Bone and tooth decay	Weak bones, tooth decay
Iodine	Physical and mental activity	Seaweed, singhada, iodised salt	Goitre	Swelling at front of neck

Nutrient	Function	Sources	Deficiency disease	Symptoms
Iron	Important part of blood	Green leafy vegetables, beetroot, pomegranate	Anaemia	Weakness, shortness of breath

DEFINITION

Nutrients are food components that provide energy, support growth, help repair and protect the body and maintain body functions: carbohydrates, proteins, fats, vitamins and minerals. Long-term lack of a nutrient causes a **deficiency disease**.

- Vitamin D can be made by our body on exposure to sunlight; salt can be obtained from rocks or seawater — not all nutrients come only from plants and animals.
- Some nutrients such as vitamin C are **lost by cooking** at high heat; cut fruits/vegetables lose some vitamins on washing, yet all should be washed thoroughly. Include fresh fruit and uncooked vegetables.

▸ Dietary fibre (roughage) and water

- **Roughage** gives no nutrients but removes undigested food and ensures smooth passage of stools (prevents constipation). Sources: green leafy vegetables, fresh fruits, whole grains, pulses, nuts.
- **Water** helps absorb nutrients and removes waste through sweat and urine; drink enough regularly.
- Locally grown, plant-based food is good for health and the planet.

▸ Know a scientist — Coluthur Gopalan (1918–2019)

Started nutrition research in India; analysed over 500 Indian foods and led surveys showing deficiencies in protein and energy. This led to the **Mid Day Meal Programme (2002)**, now **PM POSHAN**, giving balanced food in government and aided schools.

3.3 How to Test Different Components of Food?

Test for	Procedure	Positive result
Starch	Put 2–3 drops of diluted **iodine solution** on the food	**Blue-black** colour
Fat	Wrap/press food in paper, let it dry, hold against light	**Oily patch** that shines faintly through
Protein (teacher demo)	Paste in test tube + 2–3 tsp water; add 2 drops **copper sulfate** and 10 drops **caustic soda**; shake, leave a few minutes	**Violet** colour

IMPORTANT

Chemicals for the protein test are harmful: touch only when asked, wash spills with water, never taste or smell them.

- Items tested: potato, cucumber, boiled rice, boiled gram, peanuts, bread/chapati, butter, coconut (Table 3.3).
- Peanuts show **both protein and fat**, so one food can have **several nutrients**.

3.4 Balanced Diet

Nutritional needs vary with **age, gender, physical activity, health status and lifestyle**.

DEFINITION

A **balanced diet** has all essential nutrients, roughage and water in the right amounts for proper growth and development.

Per 100 g	Potato wafers	Roasted chana
Energy	536 kcal	355 kcal
Fats	35.0 g	6.26 g
Carbohydrates	53.0 g	58.58 g
Proteins	7.0 g	18.64 g
Dietary fibre	4.8 g	16.8 g

Roasted chana is the better choice. Wafers are **junk food**: high calories from sugar and fat but very little protein, minerals, vitamins and fibre; frequent use causes **obesity** and other problems. Other junk foods: candy bars, carbonated drinks.

Packaged food must list nutrients on the cover. **Fortification** adds nutrients during processing (iodised salt, some baby foods). **FSSAI** (Food Safety and Standards Authority of India) regulates food quality.

3.5 Millets: Nutrition-rich Cereals

Jowar, bajra, ragi and sanwa (barnyard millet) are small-grained **millets**, native to India, easily grown in different climates. They are rich in vitamins, minerals (iron, calcium) and fibre, so are called **nutri-cereals**.

3.6 Food Miles: From Farm to Our Plate

Wheat to chapati: farmer grows wheat, grains are threshed and winnowed, stored, ground and packed, transported to shops and reach our plate.

DEFINITION

Food miles are the entire distance travelled by a food item from producer to consumer.

- Reducing food miles cuts **cost and pollution**, supports **local farmers** and keeps food **fresh**.
- Do not waste food; take only what you can eat; respect the effort of farmers.

3.7 Likely Exam Questions

Likely question	Model answer
Why do traditional foods differ across states?	They depend on locally grown crops (soil, climate), taste, culture and traditions.
Why have cooking practices changed?	Technology, better transport and communication.
Name the main nutrients and their functions.	Carbohydrates and fats: energy; proteins: growth and repair; vitamins and minerals: protection.
What causes scurvy and goitre?	Lack of Vitamin C and lack of iodine respectively.
Write the test for starch.	Add iodine drops; blue-black colour shows starch.
How do you test for fat?	Press food on paper; an oily patch that is translucent shows fat.

Likely question	Model answer
How do you test for protein?	Add copper sulfate and caustic soda; violet colour shows protein.
What is a balanced diet?	Diet with all nutrients, roughage and water in right amounts.
Why is roughage necessary?	It helps remove undigested food and prevents constipation.
Why are junk foods unhealthy?	High sugar and fat, low protein, vitamins, minerals and fibre; cause obesity.
Why are millets called nutri-cereals?	They provide vitamins, minerals and fibre.
What are food miles and why reduce them?	Distance food travels from producer to consumer; reducing saves cost, pollution and keeps food fresh.

★ KEY TAKEAWAYS

- Food choice depends on local crops, taste, culture and tradition.
- Nutrients: carbohydrates, fats (energy), proteins (body-building), vitamins and minerals (protective).
- Deficiency causes scurvy (C), rickets (D), beriberi (B1), night blindness (A), goitre (iodine), anaemia (iron).
- Tests: iodine → blue-black (starch); paper → oily patch (fat); copper sulfate + caustic soda → violet (protein).
- Balanced diet includes roughage and water; avoid junk food.
- Millets are nutri-cereals; reduce food miles and food waste.

CHAPTER 04

Exploring Magnets

*Reshma, writing a story about a spice ship caught in a storm with no stars visible, discovers that sailors used a **magnetic compass** to find directions. This sends her exploring magnets: what they attract, their poles, how a freely hanging magnet shows north–south, how magnets interact and how to care for them.*

Big Questions

- Which materials do magnets attract?
- What are the poles of a magnet?
- How does a magnetic compass work?
- How do two magnets interact?

4.0 Magnets: Natural and Artificial

- Sailors in olden days used naturally occurring magnets called **lodestones**.
- Later people learnt that **magnets can be made from iron pieces**; today magnets are made of different materials.
- Magnets in school laboratories, pencil boxes, stickers and toys are **artificial magnets**.
- Common shapes: **bar, U-shaped (horseshoe), ring**; also disc, cylindrical and spherical.

4.1 Magnetic and Non-magnetic Materials

Activity 4.1: predict, then test with a magnet which objects (pencil, eraser, iron, plastic, wood, glass, etc.) stick. Some stick, others do not.

Type	Meaning	Examples
Magnetic materials	Attracted towards a magnet	Iron, nickel, cobalt and some of their combinations (alloys) with other metals, e.g. steel
Non-magnetic materials	Not attracted by a magnet	Wood, rubber, plastic, glass, paper, cloth

4.2 Poles of a Magnet

Activity 4.2: Spread **iron filings** on paper and place a bar magnet over them. Most filings cling near the **ends**; very few stick in the middle. This is true for magnets of any shape.

DEFINITION

The ends of a magnet, where attraction is strongest, are its two **poles**: the **North pole** and the **South pole**.

IMPORTANT

A magnet with a **single pole cannot exist**. If a magnet is broken into pieces, **each piece still has a North and a South pole** — poles always occur in pairs.

4.3 Finding Directions

Activity 4.3: Suspend a bar magnet at its middle with a thread so it balances horizontally, rotate it and let it rest, and mark the line along its ends. Repeat: it comes to rest along the **same line every time — the north–south direction**.

- The end pointing north is the **North-seeking (North) pole**; the other is the **South-seeking (South) pole**.
- This happens because the **Earth itself behaves like a giant magnet**.
- An iron bar (not a magnet) suspended the same way rests in **any** direction. So this property **tests whether a metal piece is a magnet**.
- Sun's rising/setting gives an approximate idea of east–west to locate the direction.

▸ Magnetic compass

A small circular box with a transparent cover. A **magnetised needle** is balanced on a pin so it rotates freely, with a dial marked with directions beneath it. The needle's north-pointing end is usually painted **red**. How to use: place it where directions are needed, wait for the needle to settle, then turn the box until N and S on the dial line up with the needle; now all directions are as marked on the dial.

▸ Making a magnetic compass (Activity 4.4)

Magnetise the needle
Place an iron sewing needle on a table; stroke it along its length with one pole of a bar magnet, always in the same direction, lifting the magnet at the end.

Repeat
Bring the same pole back to the same starting end and repeat 30–40 times.

Test
If iron filings or steel pins stick to the needle, it has become a magnet.

Float
Pass the needle horizontally through a cork piece and float it in a bowl of water, needle above water.

Observe
It rests north–south; rotate the cork several times — the same end always points the same way.

> **REMEMBER**
> Indians used a **matsya-yantra (machchh-yantra)** for navigation at sea: a magnetised fish-shaped iron piece floating in a vessel of oil.

4.4 Attraction and Repulsion between Magnets

Activity 4.5: Place bar magnet A on round pencils (so it can roll) and bring magnet B's ends near it without touching.

Poles brought together	Type	Result
N and S (unlike)	Unlike poles	**Attract**
N and N, or S and S (like)	Like poles	**Repel**

- With an **iron bar** replacing one magnet, both ends of the bar are attracted by both poles. Hence **repulsion is the sure test for a magnet** (attraction can happen with plain iron).
- Activity 4.6: bring the N pole of a bar magnet near the N pole of a compass needle and it **moves away**; bring the S pole near and it **moves closer** (compass needle is itself a magnet).

- Activity 4.7: placing wood, cardboard, plastic or glass between the magnet and compass causes **no appreciable change** — **magnetic effect acts through non-magnetic materials**.

4.5 Fun with Magnets

- Making a **magnetic garland**; moving objects without touching.
- Taking **steel balls out of a maze** by moving a magnet beneath a cardboard tray.
- Picking a **steel paper clip from water** with a magnet outside the water/container.
- Two matchbox cars with **like poles facing** each other move apart; with unlike poles they move together.

▸ Marking poles and keeping magnets safe

- Poles are marked **N and S**, or N shown by a white dot, or N painted red and S blue.
- Store bar magnets in **pairs with unlike poles on the same side**, with a **wooden piece** in between and **soft iron pieces across the ends**.
- Do **not heat, drop or hammer** magnets (they weaken) and keep them away from **mobile phones and remote controls**.

4.6 Likely Exam Questions

Likely question	Model answer
What are magnetic and non-magnetic materials?	Magnetic materials are attracted by a magnet (iron, nickel, cobalt); non-magnetic are not (wood, plastic, glass).
Where is the attraction of a bar magnet strongest?	Near its poles at the two ends.
Can we get a magnet with a single pole?	No. Breaking a magnet gives smaller magnets each with N and S poles.
What happens when a magnet is freely suspended?	It rests along the north–south direction because the Earth behaves like a giant magnet.
How will you identify which of two bars is a magnet?	Suspend freely or bring ends near a known magnet; repulsion/N–S alignment shows it is a magnet.
State the law of attraction and repulsion.	Like poles repel; unlike poles attract.
Why is repulsion a sure test for a magnet?	Iron is attracted to both poles, but only a magnet can repel another magnet.
Describe a magnetic compass.	A box holding a freely rotating magnetised needle over a dial with directions; it points north–south.
How can you make a magnetic compass at home?	Stroke an iron needle with a magnet 30–40 times in one direction, pass it through a cork and float it in water.
Does a magnet act through wood or glass?	Yes, magnetic effect acts through non-magnetic materials.
How should magnets be stored?	In pairs, unlike poles together, wooden piece between, soft iron across ends; avoid heat, hammering, phones.
Mechanic's steel screws keep falling. Suggest a solution.	Magnetise the screwdriver by rubbing it with a magnet so it holds the screws.

★ KEY TAKEAWAYS

- Magnets attract iron, nickel, cobalt (magnetic materials); non-magnetic materials are not attracted.
- Every magnet has two poles, N and S, which always occur in pairs.
- A freely suspended magnet rests north–south; this is the basis of the compass.
- Like poles repel, unlike poles attract; repulsion confirms a magnet.
- Magnetic force acts through non-magnetic materials.
- Store magnets properly and protect from heat, hammering and electronic devices.

CHAPTER 05

Measurement of Length and Motion

*Deepa's uniform shopping leads her and friends Anish, Hardeep, Padma and Tasneem to ask how length is measured. They learn why body-part units fail, why standard (SI) units are needed, how to read a scale correctly, how to describe position with a **reference point**, and how to classify motion as linear, circular or oscillatory.*

Big Questions

- Why do we need standard units?
- How do we measure lengths, even curved ones, correctly?
- How do we decide whether something is moving?
- What are the types of motion?

5.1 How Do We Measure?

People measure with body parts: the cloth seller uses an arm length, farmers use strides or foot lengths, a mother uses **balisht (handspan)** and tailors use **char angula** (four finger-widths). When the five friends measure their classroom table in handspans, results differ (slightly more than 13, 13, slightly less than 13, between 13 and 14, 14) because **their handspans are of different sizes**.

- A measurement has **two parts: a number and a unit** (e.g. 13 handspans).
- Units based on hand, foot, fist, finger **differ from person to person**, so a unit is needed which gives the **same result for everyone**.
- Ancient India used **angula** (finger width), multiples of angula, **dhanusa** and **yojana**; angula is still used by carpenters and tailors. Scales with ruled markings were excavated from **Harappan** sites.

5.2 Standard Units

As people travelled and traded, different local units caused confusion, so countries agreed on a common system: the **International System of Units (SI units)**.

> **DEFINITION**
> The **SI unit of length is the metre (symbol m)**.

Conversion	Value
1 kilometre (km)	1000 m
1 metre (m)	100 cm
1 centimetre (cm)	10 mm
1 millimetre (mm)	0.1 cm (smallest division on a common scale)
1 inch	2.54 cm

- On a 15 cm scale, the gap between consecutive big marks is **1 cm**, further divided into 10 equal parts of **1 mm**.

- **Km** for long distances (between cities, railway track); **cm / mm** for small lengths (page thickness, pencil).
- Inch and foot are older units still used by some people.

5.3 Correct Way of Measuring Length

- Choose a **suitable scale**: 15 cm scale for a pencil; metre scale or tape for room height; a **flexible tape** (tailor's tape) for girth of a tree or chest.
- **Place the scale in contact** with the object, along its length.
- Keep the **eye directly above** (perpendicular to) the point being read — not at a slant.
- **Broken scale / unclear zero:** start from another full mark (say 1.0 cm) and **subtract** it: length = 10.4 cm − 1.0 cm = 9.4 cm.
- Visually challenged students use scales with **raised markings**.
- Always **write the unit** and leave a **space** between number and unit.

> **NOTE**
> Unit names and symbols (km, m, cm, mm) are written in **lowercase**, have **no 's' in plural** and **no full stop** after the symbol (except at the end of a sentence). Differences between friends' readings arise from errors in positioning the scale or eye, and the limit of the scale.

5.4 Measuring the Length of a Curved Line

Use a **flexible measuring tape**, or lay a **thread** along the curve, mark its ends, straighten it and measure with a metre scale (e.g. string lights on verandah arches).

5.5 Describing Position

Each friend judged whether the garden or school was nearer **from their own house**, so each was correct for their own start point; with a common point (the bus stand) everyone agrees.

> **DEFINITION**
> A fixed object or point with respect to which a distance or position is stated is called a **reference point**.

- Drawing a **kabaddi court** with chuna powder needs a reference point on the ground.
- **Kilometre stones** ('Delhi 70 km', then 'Delhi 60 km') give position from Delhi, the reference point; the decreasing number shows Padma is getting nearer.

5.6 Moving Things

> **DEFINITION**
> An object is in **motion** if its **position changes with respect to a reference point with time**. If its position does not change, it is **at rest**.

- In the bus, passengers seated are **at rest relative to the bus (or Deepa)** but **in motion relative to buildings outside**. Hence rest and motion depend on the reference point.
- Think: on a ship moving steadily on calm sea with no window, we cannot tell whether it moves.

5.7 Types of Motion

Type	Definition	Examples / activity
Linear	Motion along a **straight line**	Dropped eraser, falling orange, pushed box, march-past, car on straight road
Circular	Motion along a **circular path**	Eraser whirled on a thread, merry-go-round, wheel rim
Oscillatory	**To-and-fro** motion about a fixed position	Swing, hanging eraser (pendulum) released from one side, vibrating metal strip fixed at one end

IMPORTANT

Motion that repeats its path after a fixed interval of time is **periodic**. **Circular and oscillatory motion are periodic.**

Rollercoaster question: the ball shows different motions on different portions (straight sections linear, loop circular, up-and-down curves not simply oscillatory unless it repeats). Always classify each portion by the path shape.

5.8 Likely Exam Questions

Likely question	Model answer
Why are body-part units not reliable?	They differ from person to person, giving different measurements of the same length.
What is the SI unit of length? Give its symbol.	Metre, m.
Write the conversions among km, m, cm and mm.	1 km = 1000 m; 1 m = 100 cm; 1 cm = 10 mm.
How will you use a scale whose end is broken?	Start at another full mark and subtract that reading from the other end's reading.
How should the eye be placed while reading a scale?	Directly above the point being read, perpendicular to the scale.
How do you measure a curved line?	Lay a thread along it, straighten it and measure it with a scale, or use a flexible tape.
What is a reference point?	A fixed point with respect to which position or distance is stated.
When is an object said to be in motion?	When its position changes with respect to a reference point with time.
Can an object be at rest and in motion at the same time?	Yes, depending on the reference point, e.g. a passenger in a bus.
Define linear, circular and oscillatory motion with examples.	Straight path (falling stone); circular path (merry-go-round); to-and-fro (swing).
What is periodic motion?	Motion that repeats after a fixed interval of time, e.g. circular and oscillatory motion.
Express 1.5 km in metres.	1.5 × 1000 = 1500 m.

★ KEY TAKEAWAYS

- A measurement = number + unit; non-standard units vary from person to person.
- SI unit of length is the metre; 1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm.
- Place scale touching the object, eye vertically above the mark; subtract readings for a broken scale.
- Curved lengths: use thread or flexible tape.
- Motion and rest are decided with respect to a reference point.
- Types of motion: linear, circular, oscillatory; the last two are periodic.

CHAPTER 06

Materials Around Us

*Ghulan and Sheeta's teacher Madam Vidya makes the class look at everyday objects: how they look, feel and differ. The chapter shows that objects are made of **materials**, that materials are chosen for their **properties**, and that they can be **classified** on appearance, hardness, transparency, solubility, mass and volume. Anything with mass and volume is **matter**.*

Big Questions

- What are objects made of and why are particular materials chosen?
- How can materials be classified?
- What properties do all materials share?
- What is matter?

6.1 Observing Objects Around Us

DEFINITION

Any substance used to make an object is a **material** (paper, wood, cloth, glass, metal, plastic, clay etc.).

- An object may be made of one material or of **several** (a pen: plastic, metal, ink); the same material can make many objects, and one object can be made from different materials.
- **Pottery in India:** earliest pottery is 7,000–8,000 years old (Lahuradewa in the Ganga plains, Mehrgarh in Baluchistan). From about 4000 BCE wheel-turned pots, coloured slips and painted designs developed. Harappan (Sindhu-Sarasvati, 2600–1900 BCE) pottery has bright red surface with black geometric and animal designs. Clay was cleaned, sieved, kneaded, wheel-turned and baked in kilns; **baked clay is terracotta**. Large jars are displayed in the National Museum, New Delhi.

6.2 How to Group Materials?

DEFINITION

Arranging objects into groups on the basis of a common property is **classification**.

- Objects can be grouped by shape, colour, hardness, shine or material.
- A material is chosen according to its **properties and the purpose** of the object: a tumbler must hold water, so paper-like cloth is unsuitable; cloth is not used for cooking utensils.
- Activity 6.4 (balls): tennis, cricket and hand-exercise balls of the same size are dropped from a fixed height; bounce differs — balls are made of specific materials for specific games.
- Classification is used in kitchens, grocer shops and chemist shops to find things easily.

6.3 Properties of Materials

▸ 6.3.1 Appearance: lustrous and non-lustrous

Type	Meaning	Examples
Lustrous	Shiny surface	Metals — iron, copper, zinc, aluminium, gold

Type	Meaning	Examples
Non-lustrous	No shiny surface	Paper, wood, rubber, jute

- Some metals lose lustre due to **air and moisture**, so lustre shows on freshly cut surfaces.
- Not all lustrous things are metals: polishing or coating with plastic/wax can make a surface shine ('all that glitters is not gold').

▸ 6.3.2 Hard and soft

Materials difficult to compress or scratch are **hard** (stone, iron, brick); those easily compressed or scratched are **soft** (eraser, sponge, candle). Hardness is **relative**: rubber is harder than sponge but softer than iron. Test by scratching with a metal key.

▸ 6.3.3 Transparency

Type	Meaning	Examples
Transparent	Objects seen **clearly** through it	Glass, water, air, cellophane, window glass
Translucent	Objects seen **but not clearly**	Butter paper, frosted glass
Opaque	Objects **cannot be seen** through it	Wood, cardboard, metals, wall, tree

▸ 6.3.4 Solubility in water

- **Soluble** — completely disappears (dissolves) when stirred in water: sugar, salt.
- **Insoluble** — does not mix or disappear: chalk powder, sand, sawdust.
- Water is vital to the body as it dissolves many materials.
- **ORS** (Oral Rehydration Solution) for dehydration: six teaspoons of sugar and half a teaspoon of salt in one litre of boiled and cooled water.
- Some liquids mix with water completely; some (oil) form a separate layer. Some gases dissolve: **oxygen dissolves in water**, which is essential for aquatic life.

▸ 6.3.5 Heavy and light — mass

Three identical cups half-filled with water, sand and pebbles are weighed on a balance; the heavier has **more mass**. Mass is the property measuring how heavy or light something is (in everyday language called 'weight').

▸ 6.3.6 Space and volume

Bags on seats leave no room to sit; the same tumbler holds half or full water depending on the amount. The **space occupied** by a material is its **volume**. Bottles marked 1 L, 500 mL show volume.

6.4 What Is Matter?

> **DEFINITION**
>
> **Matter** is anything that has **mass** and occupies **space (volume)**. Mass and volume are the two properties possessed by all materials.

Quantity	Units	How to write
Mass	gram (g), kilogram (kg); SI unit is **kg**	7 kg (not 7 kgs); no full stop; space between number and unit

Quantity	Units	How to write
Volume	litre (L), millilitre (mL); SI unit is **cubic metre (m^3)**	500 mL, 2 m^3; 1 m^3 = 1000 L

Water, sand, pebbles and a cup are all matter. Materials are the types of matter used to make objects. Air is also matter (it has mass and occupies space).

▸ Ayurveda's classification

Ayurveda describes all physical matter, living things and food using **20 properties (guna)** in **ten opposite pairs**:

Property	Opposite
guru (heavy)	laghu (light)
manda (slow)	tikshna (quick)
hima (cold)	ushna (hot)
snigdha (unctuous)	ruksha (dry)
shlakshna (smooth)	khara (rough)
sandra (solid)	drava (liquid)
mridu (soft)	kathina (hard)
sthira (stable)	khala (moving)
sukshma (subtle/small)	sthula (big)
vishada (non-slimy)	picchhila (slimy)

6.5 Likely Exam Questions

Likely question	Model answer
What is a material? Give examples.	Any substance used to make an object, e.g. wood, glass, metal, clay.
What is classification and why is it done?	Arranging objects into groups on common properties; it helps study and find them easily.
What decides the choice of material for an object?	Its properties and the purpose of the object.
Define lustre. Are all shiny materials metals?	Shiny appearance; no, polished or coated non-metals may shine.
Differentiate transparent, translucent and opaque.	Clear view, unclear view, no view through; glass, butter paper, wood.
What are soluble and insoluble materials?	Dissolve completely in water (sugar, salt) and do not dissolve (sand, chalk).
Why is hardness called a relative property?	A material may be harder than one and softer than another, e.g. rubber.

Likely question	Model answer
What is matter?	Anything that has mass and occupies space.
Give the units of mass and volume.	kg and g; m3, L and mL.
How is ORS prepared at home?	Six teaspoons of sugar and half a teaspoon salt in one litre boiled and cooled water.
Why do shops use transparent containers?	Contents can be seen without opening them.
Why is dissolved oxygen important?	Aquatic plants and animals need it to survive.

★ KEY TAKEAWAYS

- Objects are made from materials chosen for their properties and purpose.
- Classification groups materials by similarities and differences.
- Materials may be lustrous/non-lustrous, hard/soft, transparent/translucent/opaque, soluble/insoluble.
- Mass measures how heavy a material is; volume is the space it occupies.
- Anything having mass and volume is matter.
- SI units: kg for mass, m^3 for volume (L and mL are commonly used).